6.8 THE SINGLE-ENDED PRIMARY INDUCTANCE CONVERTER (SEPIC)

A converter similar to the Ćuk is the single-ended primary inductance converter (SEPIC), as shown in Fig. 6-14. The SEPIC can produce an output voltage that is either greater or less than the input but with no polarity reversal.

To derive the relationship between input and output voltages, these initial assumptions are made:

1. Both inductors are very large and the currents in them are constant.
2. Both capacitors are very large and the voltages across them are constant.
3. The circuit is operating in the steady state, meaning that voltage and current waveforms are periodic.
4. For a duty ratio of D, the switch is closed for time DT and open for $(1 - D)T$.
5. The switch and the diode are ideal.

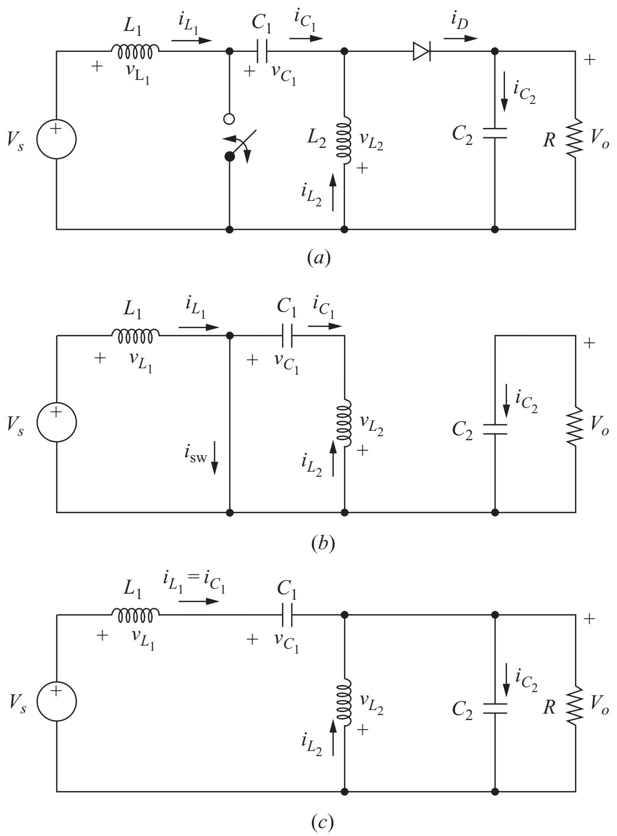

Figure 6-14 (a) SEPIC circuit; (b) Circuit with the switch closed and the diode off; (c) Circuit with the switch open and the diode on.

The inductor current and capacitor voltage restrictions will be removed later to investigate the fluctuations in currents and voltages. The inductor currents are assumed to be continuous in this analysis. Other observations are that the average inductor voltages are zero and that the average capacitor currents are zero for steady-state operation.

Kirchhoff's voltage law around the path containing V_s, L_1, C_1, and L_2 gives

$$-V_s + v_{L1} + v_{C1} - v_{L2} = 0$$

Using the average of these voltages,

$$-V_s + 0 + V_{C1} - 0 = 0$$

showing that the average voltage across the capacitor C_1 is

$$V_{C1} = V_s \tag{6-69}$$

When the switch is closed, the diode is off, and the circuit is as shown in Fig. 6-14*b*. The voltage across L_1 for the interval DT is

$$v_{L1} = V_s \tag{6-70}$$

When the switch is open, the diode is on, and the circuit is as shown in Fig. 6-14*c*. Kirchhoff's voltage law around the outermost path gives

$$-V_s + v_{L1} + v_{C1} + V_o = 0 \tag{6-71}$$

Assuming that the voltage across C_1 remains constant at its average value of V_s [Eq. (6-69)],

$$-V_s + v_{L1} + V_s + V_o = 0 \tag{6-72}$$

or

$$v_{L1} = -V_o \tag{6-73}$$

for the interval $(1 - D)T$. Since the average voltage across an inductor is zero for periodic operation, Eqs. (6-70) and (6-73) are combined to get

$$(v_{L1,\text{ sw closed}})(DT) + (v_{L1,\text{ sw open}})(1 - D)T = 0$$

$$V_s(DT) - V_o(1 - D)T = 0$$

where D is the duty ratio of the switch. The result is

$$\boxed{V_o = V_s\left(\frac{D}{1 - D}\right)} \tag{6-74}$$

which can be expressed as

$$\boxed{D = \frac{V_o}{V_o + V_s}} \tag{6-75}$$

This result is similar to that of the buck-boost and Ćuk converter equations, with the important distinction that there is no polarity reversal between input and output voltages. The ability to have an output voltage greater or less than the input with no polarity reversal makes this converter suitable for many applications.

Assuming no losses in the converter, the power supplied by the source is the same as the power absorbed by the load.

$$P_s = P_o$$

Power supplied by the dc source is voltage times the average current, and the source current is the same as the current in L_1.

$$P_s = V_s I_s = V_s I_{L1}$$

Output power can be expressed as

$$P_o = V_o I_o$$

resulting in

$$V_s I_{L1} = V_o I_o$$

Solving for average inductor current, which is also the average source current,

$$I_{L1} = I_s = \frac{V_o I_o}{V_s} = \frac{V_o^2}{V_s R} \tag{6-76}$$

The variation in i_{L1} when the switch is closed is found from

$$v_{L1} = V_s = L_1\left(\frac{di_{L1}}{dt}\right) = L_1\left(\frac{\Delta i_{L1}}{\Delta t}\right) = L_1\left(\frac{\Delta i_{L1}}{DT}\right) \tag{6-77}$$

Solving for Δi_{L1},

$$\Delta i_{L1} = \frac{V_s DT}{L_1} = \frac{V_s D}{L_1 f} \tag{6-78}$$

For L_2, the average current is determined from Kirchhoff's current law at the node where C_1, L_2, and the diode are connected.

$$i_{L2} = i_D - i_{C1}$$

Diode current is

$$i_D = i_{C2} + I_o$$

which makes

$$i_{L2} = i_{C2} + I_o - i_{C1}$$

The average current in each capacitor is zero, so the average current in L_2 is

$$I_{L2} = I_o \tag{6-79}$$

The variation in i_{L2} is determined from the circuit when the switch is closed. Using Kirchhoff's voltage law around the path of the closed switch, C_1, and L_2 with the voltage across C_1 assumed to be a constant V_s, gives

$$v_{L_2} = v_{C_1} = V_s = L_2\left(\frac{di_{L2}}{dt}\right) = L_2\left(\frac{\Delta i_{L2}}{\Delta t}\right) = L_2\left(\frac{\Delta i_{L2}}{DT}\right)$$

Solving for Δi_{L_2}

$$\Delta i_{L_2} = \frac{V_s DT}{L_2} = \frac{V_s D}{L_2 f} \tag{6-80}$$

Applications of Kirchhoff's current law show that the diode and switch currents are

$$\begin{aligned} i_D &= \begin{cases} 0 & \text{when switch is closed} \\ i_{L_1} + i_{L_2} & \text{when switch is open} \end{cases} \\ i_{sw} &= \begin{cases} i_{L_1} + i_{L_2} & \text{when switch is closed} \\ 0 & \text{when switch is open} \end{cases} \end{aligned} \tag{6-81}$$

Current waveforms are shown in Fig. 6-15.

Kirchhoff's voltage law applied to the circuit of Fig. 6-14*c*, assuming no voltage ripple across the capacitors, shows that the voltage across the switch when it is open is $V_s + V_o$. From Fig. 6-14*b*, the maximum reverse bias voltage across the diode when it is off is also $V_s + V_o$.

The output stage consisting of the diode, C_2, and the load resistor is the same as in the boost converter, so the output ripple voltage is

$$\boxed{\Delta V_o = \Delta V_{C2} = \frac{V_o D}{RC_2 f}} \tag{6-82}$$

Solving for C_2,

$$\boxed{C_2 = \frac{D}{R(\Delta V_o / V_o) f}} \tag{6-83}$$

The voltage variation in C_1 is determined form the circuit with the switch closed (Fig. 6-14*b*). Capacitor current i_{C_1} is the opposite of i_{L_2}, which has previously been determined to have an average value of I_o. From the definition of capacitance and considering the magnitude of charge,

$$\Delta V_{C_1} = \frac{\Delta Q_{C1}}{C} = \frac{I_o \Delta t}{C} = \frac{I_o DT}{C}$$

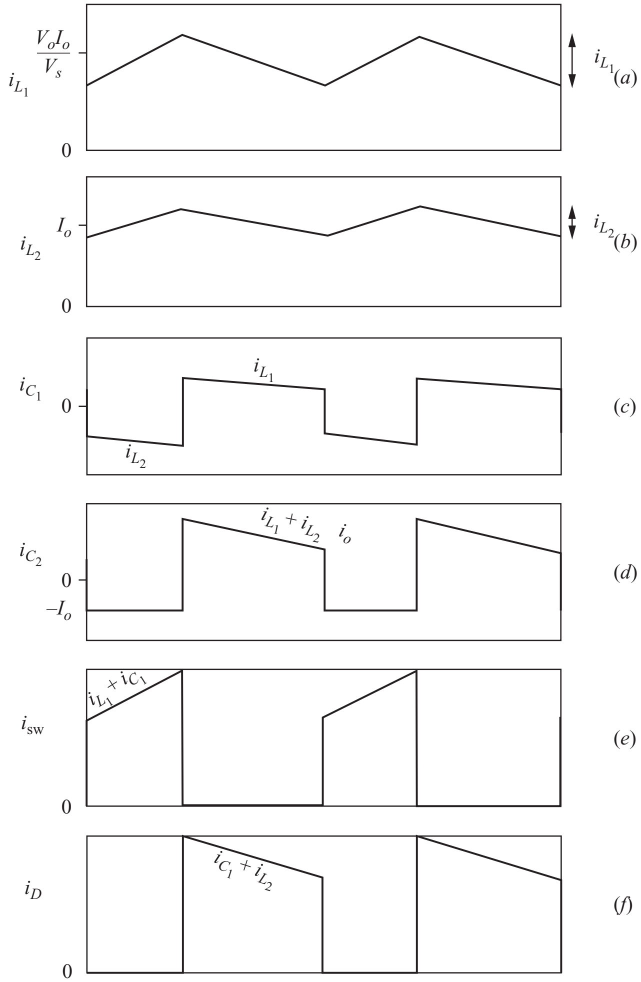

Figure 6-15 Currents in the SEPIC converter. (*a*) L_1; (*b*) L_2; (*c*) C_1; (*d*) C_2; (*e*) switch; (*f*) diode.

Replacing I_o with V_o/R,

$$\boxed{\Delta V_{C_1} = \frac{V_o D}{R C_1 f}} \tag{6-84}$$

Solving for C_1,

$$\boxed{C_1 = \frac{D}{R(\Delta V_{C_1}/V_o)f}} \tag{6-85}$$

The effect of equivalent series resistance of the capacitors on voltage variation is usually significant, and the treatment is the same as with the converters discussed previously.

EXAMPLE 6-8

SEPIC Circuit

The SEPIC circuit of Fig. 6-14*a* has the following parameters:

$$
\begin{aligned}
V_s &= 9\text{ V}\\
D &= 0.4\\
f &= 100\text{ kHz}\\
L_1 = L_2 &= 90\ \mu\text{H}\\
C_1 = C_2 &= 80\ \mu\text{F}\\
I_o &= 2\text{ A}
\end{aligned}
$$

Determine the output voltage; the average, maximum, and minimum inductor currents; and the variation in voltage across each capacitor.

■ **Solution**

The output voltage is determined from Eq. (6-74).

$$V_o = V_s\left(\frac{D}{1-D}\right) = 9\left(\frac{0.4}{1-0.4}\right) = 6\text{ V}$$

The average current in L_1 is determined from Eq. (6-76).

$$I_{L1} = \frac{V_o I_o}{V_s} = \frac{6(2)}{9} = 1.33\text{ A}$$

From Eq. (6-78)

$$\Delta i_{L1} = \frac{V_s D}{L_1 f} = \frac{9(0.4)}{90(10)^{-6}(100{,}000)} = 0.4\text{ A}$$

Maximum and minimum currents in L_1 are then

$$I_{L1,\max} = I_{L1} + \frac{\Delta i_{L1}}{2} = 1.33 + \frac{0.4}{2} = 1.53\text{ A}$$

$$I_{L1,\min} = I_{L1} - \frac{\Delta i_{L1}}{2} = 1.33 - \frac{0.4}{2} = 1.13\text{ A}$$

For the current in L_2, the average is the same as the output current $I_o = 2$ A. The variation in I_{L_2} is determined from Eq. (6-80)

$$\Delta i_{L2} = \frac{V_s D}{L_2 f} = \frac{9(0.4)}{90(10)^{-6}(100{,}000)} = 0.4\text{ A}$$

resulting in maximum and minimum current magnitudes of

$$I_{L2,\max} = 2 + \frac{0.4}{2} = 2.2\text{ A}$$

$$I_{L2,\min} = 2 - \frac{0.4}{2} = 1.8\text{ A}$$

Using an equivalent load resistance of 6 V/2 A = 3 Ω, the ripple voltages in the capacitors are determined from Eqs. (6-82) and (6-84).

$$\Delta V_o = \Delta V_{C_2} = \frac{V_o D}{R C_2 f} = \frac{6(0.4)}{(3)80(10)^{-6}(100{,}000)} = 0.1 \text{ V}$$

$$\Delta V_{C_1} = \frac{V_o D}{R C_1 f} = \frac{6(0.4)}{(3)80(10)^{-6}(100{,}000)} = 0.1 \text{ V}$$

In Example 6-8, the values of L_1 and L_2 are equal, which is not a requirement. However, when they are equal, the rates of change in the inductor currents are identical [Eqs. (6-78) and (6-80)]. The two inductors may then be wound on the same core, making a 1:1 transformer. Figure 6-16 shows an alternative representation of the SEPIC converter.

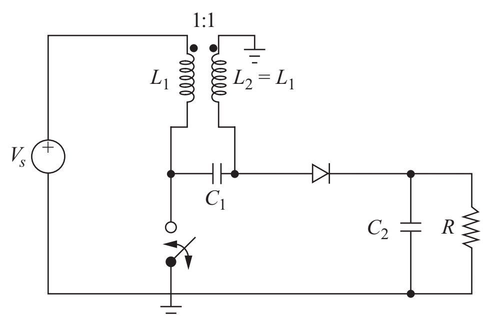

Figure 6-16 A SEPIC circuit using mutually coupled inductors.

6.9 INTERLEAVED CONVERTERS

Interleaving, also called *multiphasing*, is a technique that is useful for reducing the size of filter components. An interleaved buck converter is shown in Fig. 6-17*a*. This is equivalent to a parallel combination of two sets of switches, diodes, and inductors connected to a common filter capacitor and load. The switches are operated 180° out of phase, producing inductor currents that are also 180° out of phase. The current entering the capacitor and load resistance is the sum of the inductor currents, which has a smaller peak-to-peak variation and a frequency twice as large as individual inductor currents. This results in a smaller peak-to-peak variation in capacitor current than would be achieved with a single buck converter, requiring less capacitance for the same output ripple voltage. The variation in current coming from the source is also reduced. Figure 6-17*b* shows the current waveforms.

The output voltage is obtained by taking Kirchhoff's voltage law around either path containing the voltage source, a switch, an inductor, and the output voltage. The voltage across the inductor is $V_s - V_o$ with the switch closed and